

A 3D-Printed Series-Elastic Actuator for a Low-Cost Upper-Limb Rehabilitation Exoskeleton

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TRACK T2 · ROBOTICS, DRONES & AUTONOMOUS SYSTEMS

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1 PROBLEM STATEMENT

Post-stroke rehabilitation needs an actuator that yields when the patient's arm resists; a rigid geared joint fights the limb it is meant to assist and cannot sense interaction torque. Commercial series-elastic actuators solve this at ₹1.8–2.5 lakh per joint, so a district hospital equips one bed rather than a ward.

2 OBJECTIVES

- Design a four-spoke planar torsion spring printable in PETG, target stiffness 4.5 N·m/rad.
- Characterise stiffness and hysteresis over 20,000 loading cycles at ± 3 N·m.
- Achieve torque sensing better than 0.05 N·m from encoder deflection alone.
- Validate closed-loop torque tracking on an elbow joint against a reference load cell.

3 METHODOLOGY

- Spring design — FEA sweep over 14 spoke-root radii, PETG, 40% infill
- Print and assembly — 0.15 mm layers, annealed at 70 °C for four hours
- Static characterisation — dead-weight loading 0–6 N·m, deflection logged
- Fatigue testing — 20,000 cycles at ± 3 N·m, stiffness drift tracked
- Closed-loop trials — PI torque control at 200 Hz against a load cell

4 EXPERIMENTAL SETUP

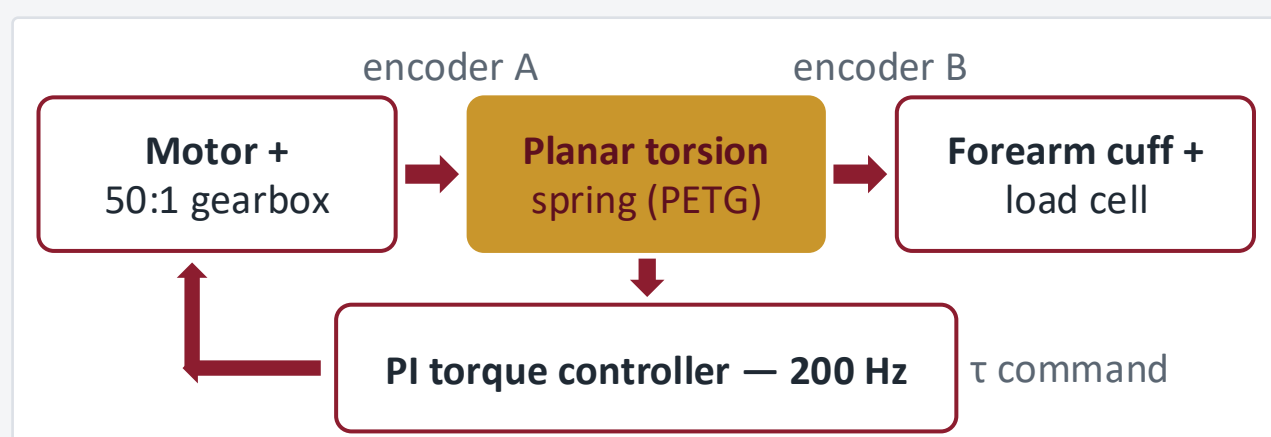


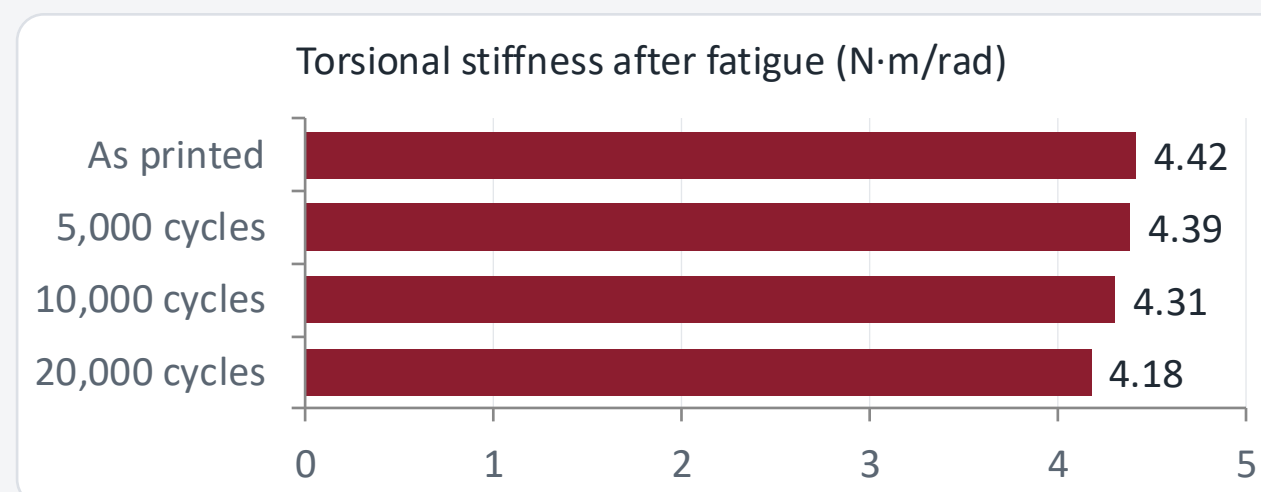
Fig. 1 — Actuator test bench. Encoders on either side of the printed torsion spring give the deflection; the PI loop closes torque on that deflection and commands the motor.

5 RESULTS & DISCUSSION

4.42
MEASURED STIFFNESS
(N·m/rad)

0.03 N·m
TORQUE SENSING
RESOLUTION

₹9,400
MATERIAL COST
PER JOINT



Stiffness falls 5.4% over 20,000 cycles, almost all of it within the first 5,000 as the printed layers bed in; annealing before calibration removes most of that drift. Hysteresis stays under 4% of full scale. The spring failed at 31,000 cycles by spoke-root cracking, which sets the service interval.

6 NOVELTY & OUR OWN CONTRIBUTION

- Four-spoke geometry sized by my own FEA sweep over 14 spoke-root radii; the 3.2 mm fillet is the only value that survived fatigue.
- Full 20,000-cycle fatigue characterisation of a PETG torsion spring — no published data existed for this material in this role.
- Annealing and pre-calibration bedding-in procedure devised by me and shown to remove most of the stiffness drift.
- Reused and cited: the series-elastic actuator concept [1], impedance control formulation [3] and standard hysteresis metrics [4].

7 CONCLUSION & IMPACT

A ₹9,400 printed series-elastic actuator delivers 4.42 N·m/rad with 0.03 N·m torque resolution — sufficient for elbow rehabilitation — against ₹1.8 lakh for a commercial unit. A district hospital could equip a ward rather than a single bed. Next: fatigue life beyond 30,000 cycles and a two-degree-of-freedom shoulder module.

SDG 3

SDG 9

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8 REFERENCES (IEEE style, max 6)

- [1] G. A. Pratt and M. M. Williamson, "Series elastic actuators," Proc. IEEE/RSJ IROS, pp. 399–406, 1995.
- [2] J. E. Pratt and B. T. Krupp, "Series elastic actuators for legged robots," Proc. SPIE, vol. 5422, pp. 135–144, 2004.
- [3] N. Hogan, "Impedance control: an approach to manipulation," ASME J. Dyn. Syst. Meas. Control, vol. 107, pp. 1–24, 1985.
- [4] H. Vallery et al., "Compliant actuation of rehabilitation robots," IEEE Robot. Autom. Mag., vol. 15, no. 3, pp. 60–69, 2008.